

Computing Ethics, Philosophy of Ethics, Professionalism

Lecture 3

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Rapid Pace of Change

- 1940s: The first computer is built
- 1956: First hard-disk drive weighed a ton and stored five megabytes
- 1991: Space shuttle had a one-megahertz computer
- 2006: Pocket devices hold a terabyte (one trillion bytes) of data
- 2006: Automobiles can have 100-megahertz computers

Rapid Pace of Change:

Discussion Question

- What devices are now computerized that were not originally? Think back 10, 20, 50 years ago.
- With change in technology, what are the Social impacts and controversies morph constantly.

Rapid Pace of Change

In addition to automobiles the following:

- TVs
- Clocks and watches
- Phones
- Cash registers
- Cameras
- Ovens
- Etc.

New Developments

Blogs (Word made up from 'web log'):

- Began as outlets for amateurs who want to express ideas or creativity
- Appealing because present personal views, are funny and creative, and present a quirky perspective on current events

New Developments (cont.)

Blogs (cont.):

- Now used as alternatives to mainstream news and for business public relations
- Popular blogs have 100,000 to 500,000 readers per day and can peak at several million views per day

New Developments (cont.)

Video Sharing:

- Rise of amateur videos on the web
- Boom of websites like Youtube and Myspace
- Many videos on the web can infringe(break) copyrights owned by entertainment companies

New Developments (cont.)

Kill Switch

- In 2011, discovered malicious code in an app for Android phones, Google quickly removed
- What are some potential uses and risks?
- What if malicious hackers found a way to operate the kill switches on our devices?
- Will the impact of electronic kill switches be more devastating than attempts to prohibit printed material?

New Developments (cont.)

Social Networking:

- First online social networking site was www.classmates.com in 1995
- MySpace, founded in 2003 had roughly 100 million member profiles by 2006
- Facebook was started at Harvard as an online version of student directories
- Crowd-funding: Making funding via internet
- How do social networking sites affect people and relationships?
- Does the time spent online reduce the time spent on physical activity and staying healthy?

Examples

- Sites like Second Life (www.secondlife.com) combine many of the features of social networking sites with the 3-D aspects of video games. What new problems/benefits arise when a person can take on a physical persona (an avatar) that may be completely different from who they are in real life? Some people with physical disabilities can interact with others without revealing their handicap.

New Developments (cont.)

Collaboration:

- Wikipedia, the online, collaborative encyclopedia
- Collaboration between scientists in different states or countries
- Watch-dogs on the Web

Examples

- Wikipedia's reliability was brought into question when a major contributor was found to be a 14 year-old boy instead of a scientist with a PhD and years of experience. He got most of his information from other Web sources.
- Web Watch-dog sites are controversial. A web site where a woman can publish the names and pictures of men who other women should watch out for has made the news recently.

New Developments (cont.)

E-commerce and Free Stuff:

- Free stuff on the web: email, books, newspapers, games, etc.
- www.Amazon.com started in 1994 and 10 years later annual sales reached \$8.5 billion
- TV show episodes are available to view on the Web

New Developments (cont.)

Artificial Intelligence (AI), Robotics, and Motion:

- AI suited to narrow, specialized skills
- Robotic devices often special-purpose devices, and may require AI to function
- Motion sensing devices are used to give robots the ability to walk, trigger airbags in a crash and cushion laptops when dropped
- E.g. personal Assistants

New Developments (cont.)

Tools for Disabled People:

- Restoration of abilities, productivity and independence
- Screen readers and scanners
- Speech recognition
- Prosthetics and motion sensors

New Developments (cont.)

Discussion Question

- What changes and new developments do you expect in the next 50 years?
- How will life be different than it is today?

Examples

Some ideas to add to the discussion:

- Wearable computers
- Enhanced intelligence (via instant communications and/or direct computer interface)

Issues and Themes

Issues:

- Unemployment
- Crime
- Loss of privacy

Issues and Themes (cont.)

Themes:

- Old problems in a new context: crime, violent fiction, and games, advertising, copyright infringement, gambling, and products
- Adapting to new technology: require adaptive changes in laws, social institutions, business policies, and personal skills, attitudes, and behavior
- Varied sources of solutions to problems: natural part of change and life

Issues and Themes (cont.)

Themes (cont.):

- Perfection is a direction, not an option.
- Trade-offs and controversy: increasing security means reducing convenience
- Difference between personal choices, business policies, and law

Ethics

What is Ethics:

- Study of what it means to “do the right thing”
- Ethical rules as fundamental and universal
- Rules to follow in our interactions and our actions that affect others
- “Is it right to . . . ?” Is it right to make a significant change in your company’s privacy policy without giving customers or members advance notice?

Natural Rights

- Respects a set of fundamental rights of others, including the rights to life, liberty, and property.
- Increasing people's happiness

Important Distinctions

- Right, wrong and okay
- **Negative rights (liberties)**
 - The right to act without interference
 - E.g. “life, liberty, and the pursuit of happiness”
- **Positive rights (claim-rights)**
 - An obligation of some people to provide certain things for others
 - E.g. internet access or a job

Contributing to Society

- Aristotle says that one lives a virtuous life by doing virtuous acts.
- how to live a virtuous life? E.g. Charity work
- Doing one's work honestly, responsibly, ethically, creatively, and well is a virtuous activity.

Social contracts and a theory of political justice

- A just and fair political system will ensure that all citizens have sufficient means to make effective use of their freedoms
- An action or a social or political structure is not ethical if it has the effect of leaving the least-advantaged people worse than they were before
- Rawls proposed “Veil of ignorance”

Veil of Ignorance

- The **Veil of Ignorance**, a component of social contract theory, allows us to test ideas for fairness. Behind the **Veil of Ignorance**, no one knows who they are. They lack clues as to their class, their privileges, their disadvantages, or even their personality.

Computer Ethics

Rationale for studying computer Ethics

- Technology is entrenched in our society, ethical questions arise regarding the uses of this new technology
- Are these ethical issues pertaining to computer professionals new or are these the same issues in a different guise?

❑ **Profession**: A paid occupation, especially one that involves prolonged training and a formal qualification.

❑ **Professional**: A professional is a member of a profession or any person who earns their living from a specified professional activity. The term also describes the standards of education and training that prepare members of the profession with the particular knowledge and skills necessary to perform their specific role within that profession.



Professional Responsibilities

With reference to Information Technology, Computer Science or Software Engineering, the responsibilities of working professionals in this area include network administration, software development and installation, and the planning and management of an organization's technology life cycle, by which hardware and software is maintained, upgraded and replaced.

- But this is **not Enough....**

The Professionalism

- ❑ A profession isn't just what you do, it's who you are.
- ❑ Professionalism is a way of thinking and living rather than an accumulation of learning.

A professional behaves ethically

- ❑ Ethics means something more than 'law' and 'morals'
 - ❑ It carries an additional connotation of 'rightness'.
 - Breaking the law: can earn a fine or jail time
 - Breaking a moral: can ruin your reputation
 - Breaking an ethic: can ruin your conscience
- It's possible to break all three, simultaneously!

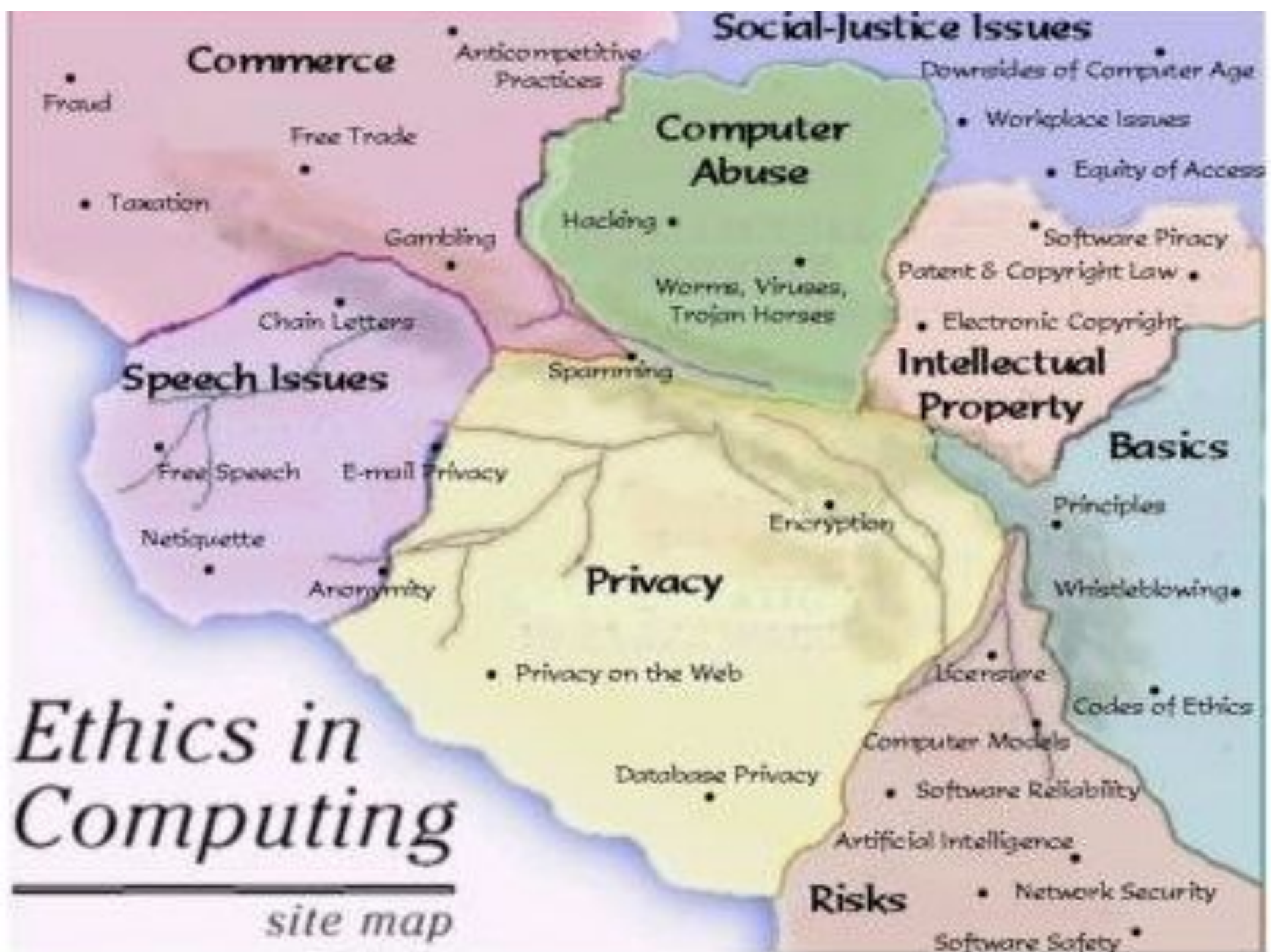
Categories of Computer Issues

- **Privacy**
 - Computers create a false sense of security
 - People do not realize how vulnerable information stored on computers are
- **Property**
 - Physical property
 - Intellectual property (in both copyright and patent)
 - Data as property
- **Access**
 - Access to computing technology
 - Access to data
- **Accuracy**
 - Accuracy of information stored



Moral and Ethical Problems

- **Hacking, cracking and virus creation**
 - Serious crimes that cannot be justified
 - Attempts at justifying such actions
 - Electrons are free- they do not belong to anybody
 - Companies have weak protection
 - Point out flaws and vulnerabilities in information systems
 - Hacking or virus creation is right in a particular country or culture
- **Software piracy**
 - Unauthorized copying of software is stealing
 - It is morally wrong as it constitutes a violation of someone else's rights



Ethical Scenarios

- Using a software trial version for 12 months, saying that s/he is evaluating it before deciding to buy it. Whether such conduct is ethical?
- A catastrophic bug has been identified just a day before deploying a safety critical software. The Q.A team has been warned not to report the bug to avoid delay in deploying software. Is it ethical?

Ethical Scenarios (contd.)

- A software that can predict the radiation of a nuclear system up to 75% accuracy, is to be developed, the software has acquired 74.6% accuracy. The manager has decided to round it off to meet client's requirement. Is it ethical?

James Moor

- Professionals face new ethical questions because the use of computer technology
- "Vacuum of rules or policies" leaves these computer professions with no guidance
- Advocates a logical conceptual framework within which to formulate a policy for computer ethics

Invisibility factor

- Invisible abuse

Invisible operations of the computer are used for unethical conduct

- Invisible programming values

Programmer values are put into the program which may include a programmer's bias

- Invisible complex calculation

Computers can perform a tremendous number of calculations

John Barlow

- **Cyberspace** - a vast continent with no end to its dimensions
- Need for "new social contracts for the digital domain"
- Concern for: privacy issues, freedom of expression, economic opportunity, and property rights, cultural implications of the technology

Ethics in Society

- Society

- An association of people organized under a system of rules designed to advance the good of its members over time
- **Morality:** Rules of conduct describing what people must and must not to do
- **Ethics:** Philosophical study of morality, a rational examination into people's moral belief's and behavior